

fonctions

a) $x + y = 100$

↓
 $y = 100 - x \rightarrow$ équation de pente -1
 et OO de $O = 100$
 $(0; 100)$
 $\hookrightarrow$ fonction

b) $x - 3 = y^2$

$x = 4 \rightarrow 4 - 3 = y^2$
 $\downarrow$
 $1 = y^2$
 $\pm 1 = y$
 $\hookrightarrow$ relation

c) $y = \sqrt{3-x} \rightarrow$ fonction

e) $x + y^3 = 27$

$x = 0 \rightarrow y^3 = 27$
 $\downarrow$
 $\sqrt[3]{y^3} = \sqrt[3]{27}$
 $\downarrow$
 $y = 3$
 $y = \sqrt[3]{-x+27}$

f) $x^3 + 1 = y^2$
 $x = 3 \rightarrow 3^3 + 1 = y^2$
 $\downarrow$
 $\sqrt{28} = \sqrt{y^2}$
 $\downarrow$
 $\pm \sqrt{28} = y$

a) Domaine $\rightarrow \text{Dom}(f) = \{x \in \mathbb{R} \mid f(x) \text{ est définie}\}$

$= \mathbb{R}$

Image $\rightarrow \text{Im}(f) = \mathbb{R}$

c) $\text{Dom}(f) = \{x \in \mathbb{R} \mid f(x) \text{ est déf}\}$

$3 - x \geq 0$

$3 \geq 0$

$=]-\infty, 3]$

$\text{Im}(f) = \{y \in \mathbb{R} \mid y \in f(x)\}$

$=]-\infty, 0]$

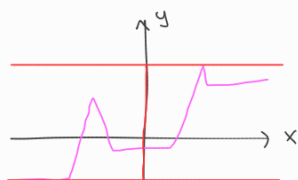
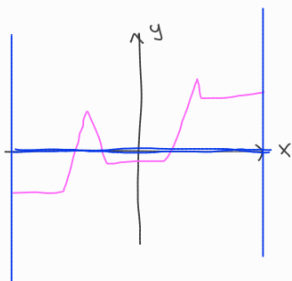


Image $\rightarrow$

$\hookrightarrow$ tout les valeurs / interval
de y de la fonction

Domaine $\rightarrow$

$\hookrightarrow$ tout les
valeurs / intervalles
de x de la
fonction



#5.19 P.31

e) $(g+f)(x) = g(x) + f(x) = 3x^2 - x + 1 + 3x + 2 = 3x^2 + 2x + 3$

b) $(g-f)(-1) = g(-1) - f(-1) = (3(-1)^2 - (-1) + 1) - (3(-1) + 2) = 6$

h) $(f \div g)(x) = \frac{f(x)}{g(x)} = \frac{3x+2}{3x^2-(x)+1}$

#5.20 P.31

L) $(f \circ g)(3) \rightarrow f(g(3)) \rightarrow f(-2) \rightarrow 1$

K) $(g \div f)(3) \rightarrow \frac{g(3)}{f(3)} \rightarrow \frac{-2}{1} = -2$

j) $(f \cdot g)(2) \rightarrow f(2) \cdot g(2) \rightarrow$ impossible $2 \notin \text{Dom}(f)$

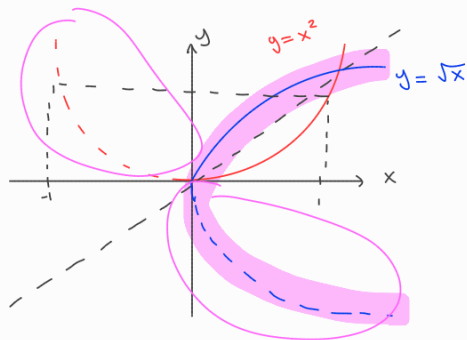
i) $(g-f)(2) \rightarrow g(2) - f(2) \rightarrow (-4) - (1) \rightarrow -5$

Récaproque

$$f(x) = y = x^2$$

$$\text{Dom}(f) = \mathbb{R}^+$$

$$\text{Im}(f) = \mathbb{R}^+$$



■ = $\emptyset$ fonction

à retirer

$$y = x^2$$

$$x = y^2$$

$$+ \sqrt{x} = y = f^{-1}(x) \quad \text{Dom}(f^{-1}) = \mathbb{R}^+$$

$$\text{Im}(f^{-1}) = \mathbb{R}^+$$

$$(f^{-1} \circ f)(x) = f^{-1}(f(x)) = f^{-1}(x^2) = \sqrt{x^2} = |x| = x \quad \text{car } x \geq 0$$

$$(f \circ f^{-1})(x) = f(f^{-1}(x)) = f(\sqrt{x}) = (\sqrt{x})^2 = x$$